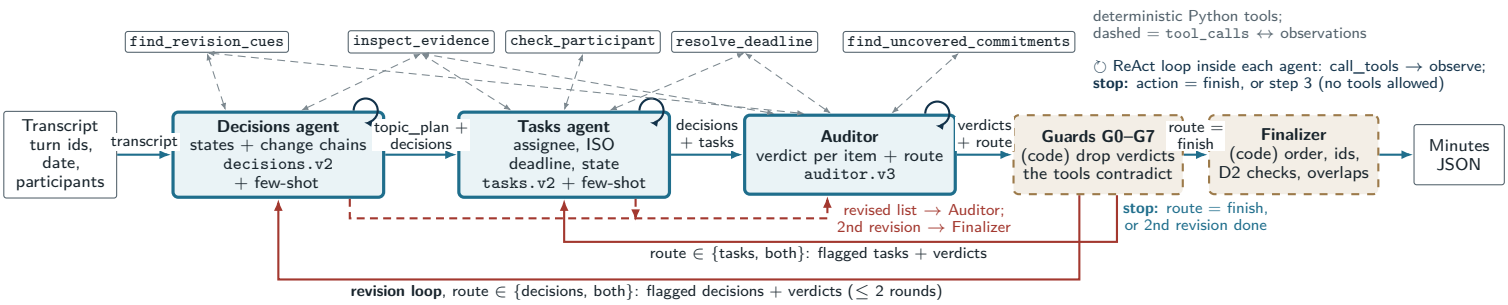
Verifiable meeting minutes with three agents that decide

Deliverable 3

Benjamin Alvear ▪ Xavier Godoy ▪ Eduardo Ruiz ▪ Damian Vera

Generative Artificial Intelligence (580694) ▪ 31-oct-2026 ▪ repository on GitHub ▪ video (M01 demo)



1. PROBLEM AND WHY THE BENCHMARK IS HARD

A Spanish meeting transcript becomes JSON minutes: each **decision** has a state (`final`, `superseded`, `rejected`, `pending`) and points to the decision it replaces; each **task** has an assignee, an ISO deadline and a condition; every item cites its evidence turn. D2 (Ministral 3B, constrained JSON + deterministic tools, two fixed passes) left four errors (D2 report, same 10 test meetings): **E1** non-final decision states (10/30 found); **E2** relative deadlines (12/39 right); **E3** confirmed tasks emitted as `pending` (47/56); **E4** nothing checks that the cited turn supports the item.

Test set: M01–M10, natural dialogue, 347 turns	n	Why it is hard
Decisions	56	only 26 are a plain `final`: 11 superseded, 11 rejected, 8 pending
Topics revised during the meeting	20	chains of up to 3 proposals
Conditional decisions / tasks	10 / 15	keep the condition, do not drop the item
Tasks	53	29 relative deadlines, 8 with a time, 22 must stay `null`
Mentions of absent people	24	they cannot be assignees

2. AGENT DESIGN

Agent	Receives → returns	Decides; stops
Decisions	transcript (+ flagged items) → `topic_plan`, decisions	call tools or finish, per step; stops at `finish` or step 3
Tasks	transcript + decisions → tasks	same ReAct loop, own tools
Auditor	transcript + both lists → verdicts, route	`finish`, `decisions`, `tasks` or `both`; ≤2 rounds

Why this split. Decisions and tasks fail differently (E1 vs E2/E3) and need different tools, so each specialist gets a short, focused prompt, which matters for a 3B model. Only the Auditor sees the whole minutes, so it covers E4 and omissions across lists. Guards and Finalizer are code and never call the model. **The agents do decide:** on test they made 212 tool calls; 3 meetings closed after the first audit and 7 ran two revision rounds; the guards rejected 41 of the Auditor's 70 flags (e.g. a `wrong_state` on a turn that literally says "pendiente") and 20 duplicate items.

3. PROMPTING TECHNIQUES (THE PROMPT REPORT TAXONOMY)

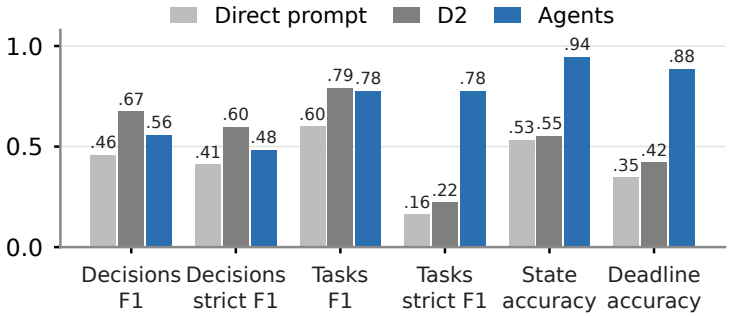
Technique (category)	Where	Fixes	ΔF1 dec.	tasks
ReAct tool use (Agents)	all	E2, E3, E4	−.031	−.051
Chain-of-Verification + Self-Refine loop (Self-Criticism)	Auditor	E4	−.004	+.025
1 train exemplar (Few-Shot ICL)	Dec., Tasks	E1, E3	+.027	−.065
`thought`, `topic_plan` (Thought Gen. / CoT)	all	E1	−.023	+.017
Guards on Auditor verdicts (code)	Auditor	E4 (verdicts)	−.068	+.005

Δ = condition without the technique minus the full system. *Declared, not ablated:* role prompting, decomposition into specialists, constrained JSON output.

4. DEVIATIONS AND LIMITATIONS

- Same task, failure, model, test inputs and evaluator as D2. Train/validation are templated ("Hablante 0X"); test is natural dialogue. Guards, `auditor.v3` and the Finalizer's overlap rule were designed on validation only (agents beat D2 on strict F1 in 10/10 validation meetings); test was run once, at commit d36043b. Direct prompt and D2 were re-run with their own prompts in the same harness and Ollama build; D2 scores .673/.789 there vs .667/.716 in the D2 report, so we compare against the stronger re-run.
- Train/validation rubrics encode "antes del viernes" as date+3 and cite superseded decisions at the replacement turn (test cites the proposal); prompts v2 follow the test convention.
- Not ablated: role prompting, constrained decoding; `no_cot` removes output-side CoT only. The last revision is not re-audited.

5. EVIDENCE: SAME 10 TEST MEETINGS, SAME D2 EVALUATOR



Condition	dec. F1	tasks F1	strict D	strict T	state	deadl.	time (s)
Direct prompt	.458	.598	.410	.161	.533	.346	180
D2	**.673**	.789	**.596**	.220	.551	.419	250
Agents (full)	.556	.775	.481	.775	.945	.884	1358
− ReAct	.525	.724	.424	.724	.918	.833	1316
− Auditor	.552	**.800**	.476	**.800**	.945	.909	484
− Few-shot	.583	.710	.450	.710	.890	.868	1646
− CoT	.533	.792	.450	.792	.932	.929	676
− Guards	.488	.780	.439	.780	.961	.891	1714

An item matches when its text similarity to the rubric is ≥ 0.55 (D2 criterion); strict F1 also requires the right state; state and deadline accuracy are over matched items; time = summed inference per step on a T4.

- **Tasks, states and deadlines improve a lot:** tasks strict F1 .22→.78 (D2 left 44 tasks `pending`, the agents none); correct deadlines 42%→88%, invented ones 11→3. Per meeting, the agents beat D2 on strict F1 in **7/10**.
- **Decisions regress** (F1 .673→.556): more rejections found (6/11 vs 2/11), fewer pending (1/8 vs 5/8) and conditional ones (2/10 vs 8/10).
- **The ablations explain it.** Without the few-shot, decisions F1 rises to .583 and 5/8 pending reappear: the exemplar comes from templated train meetings and pushes terse statements (7 words vs 16 in D2) that the similarity matcher penalizes. The guards are the largest single help for decisions (−.068 without them). The Auditor loop costs 2.8× the time for ±.03.
- **Post-hoc check** (replaying the traces, no model calls): dropping the Finalizer's decision/task overlap rule recovers 4 correct decisions but adds 10 wrong ones (.556→.557), so the regression comes from the Decisions agent.

6. FAILURE CASE: M09 (WORST STRICT F1, CRITERION FIXED BEFOREHAND)

Strict F1 .50, decisions F1 .33 (trace `runs/test/traces/full/M09.jsonl`). The chain "Friday → Monday 16" collapsed into one `final` at u16. Tasks turned u17 ("Queda fijada la primera prueba") into a task, so the Finalizer dropped the correct decision on that turn. Turns that repeat a commitment (u09/u10, u12) became duplicate tasks, and the pending "future activation" (u30) was merged into a conditional decision. The Decisions agent is where the next iteration should go: longer statements and one chain per topic.

7. REPRODUCIBILITY

The graph is a LangGraph `StateGraph`; every agent uses Ministral 3B Instruct `q4_K_M` (digest `f04aa1c7`) on Ollama 0.40.11, temperature 0, seed 42. Each prompt is a versioned file in `deliverable3/prompts/` with a CHANGELOG. Traces of every input are in `runs/test/traces/`; the video run of M01 (`runs/video/`) reproduces the evaluated trace event by event. In the Colab notebook, `MODE='DEMO'` reproduces the video and `'FULL'` the table.